

**DYNAMICAL SYSTEMS:
CLASS AUGUST 27, 2026
3.A NONDIMENSIONALIZATION.**

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NONDIMENSIONALIZATION

Problem 1. Let

$$\dot{N} = rN \left(1 - \frac{N}{K} \right) - H.$$

This is a logistic population model where there is a constant harvesting rate reducing population growth.

- (a) For each of the variables and each of the parameters, identify its associated dimension. Write this out in expressions of the form $[a] = L$.

Solution. Let P denote the dimension of population and let T denote the dimension of time. Since N is a population and t is time,

$$[N] = P, \quad [t] = T.$$

The left-hand side has dimension

$$[\dot{N}] = \left[\frac{dN}{dt} \right] = \frac{P}{T}.$$

The logistic term must therefore also have dimension P/T :

$$\left[rN \left(1 - \frac{N}{K} \right) \right] = \frac{P}{T}.$$

The quantity inside the parentheses must be dimensionless. Therefore,

$$\left[\frac{N}{K} \right] = 1,$$

so

$$[K] = P.$$

It follows that

$$[r][N] = \frac{P}{T},$$

and hence

$$[r] = T^{-1}.$$

Finally, the harvesting term H must have the same dimension as $\dot{N}$, so

$$[H] = \frac{P}{T}.$$

Thus,

$$\boxed{[N] = P, \quad [t] = T, \quad [r] = T^{-1}, \quad [K] = P, \quad [H] = PT^{-1}.}$$

- (b) Create dimensional constants N_0 and T_0 , and use them to create nondimensional variables

$$x = \frac{N}{N_0} \quad \text{and} \quad \tau = \frac{t}{T_0}.$$

Substitute x and τ into the equation and simplify.

Solution. Define

$$x = \frac{N}{N_0}, \quad \tau = \frac{t}{T_0}.$$

Thus,

$$N = N_0 x, \quad t = T_0 \tau.$$

We transform the derivative using the chain rule:

$$\frac{dN}{dt} = \frac{d(N_0 x)}{d\tau} \frac{d\tau}{dt}.$$

Since

$$\frac{d\tau}{dt} = \frac{1}{T_0},$$

we obtain

$$\frac{dN}{dt} = \frac{N_0}{T_0} \frac{dx}{d\tau}.$$

Substituting $N = N_0 x$ into the original equation gives

$$\frac{N_0}{T_0} \frac{dx}{d\tau} = r N_0 x \left(1 - \frac{N_0 x}{K} \right) - H.$$

Multiplying through by T_0/N_0 ,

$$\frac{dx}{d\tau} = r T_0 x \left(1 - \frac{N_0}{K} x \right) - \frac{H T_0}{N_0}.$$

Therefore,

$$\boxed{\frac{dx}{d\tau} = r T_0 x \left(1 - \frac{N_0}{K} x \right) - \frac{H T_0}{N_0}.}$$

Equivalently, expanding the logistic term,

$$\boxed{\frac{dx}{d\tau} = r T_0 x - \frac{r T_0 N_0}{K} x^2 - \frac{H T_0}{N_0}.}$$

- (c) List all nondimensional groups that arise. Consider a combination a nondimensional group if the combination is nondimensional but every piece would be dimensional if it were broken apart in any way.

Solution. From

$$\frac{dx}{d\tau} = rT_0x \left(1 - \frac{N_0}{K}x\right) - \frac{HT_0}{N_0},$$

the nondimensional groups are

$$rT_0, \quad \frac{N_0}{K}, \quad \frac{HT_0}{N_0}.$$

We may verify each directly.

First,

$$[rT_0] = T^{-1}T = 1.$$

Second,

$$\left[\frac{N_0}{K}\right] = \frac{P}{P} = 1.$$

Third,

$$\left[\frac{HT_0}{N_0}\right] = \frac{(P/T)T}{P} = 1.$$

Thus,

$$\boxed{rT_0, \quad \frac{N_0}{K}, \quad \frac{HT_0}{N_0}}$$

are the nondimensional groups that arise.

- (d) Make choices for values of the constants T_0 and N_0 that eliminate two of the nondimensional groups.

Solution. A natural choice is to make

$$rT_0 = 1.$$

Therefore choose

$$\boxed{T_0 = \frac{1}{r}}.$$

We can also choose

$$\frac{N_0}{K} = 1,$$

which gives

$$\boxed{N_0 = K}.$$

With these choices,

$$rT_0 = 1 \quad \text{and} \quad \frac{N_0}{K} = 1.$$

The differential equation becomes

$$\frac{dx}{d\tau} = x(1 - x) - \frac{H}{rK}.$$

- (e) Define a new nondimensional parameter (use a Greek letter such as $\alpha, \beta, \gamma, \mu$), and rewrite your equation as a nondimensional one.

Solution. Define

$$\alpha = \frac{H}{rK}.$$

The nondimensional equation becomes

$$\frac{dx}{d\tau} = x(1 - x) - \alpha.$$

Here

$$x = \frac{N}{K}, \quad \tau = rt, \quad \alpha = \frac{H}{rK}.$$

- (f) How many parameters are there in the nondimensional system? How does this compare to the number in the dimensional version? Notice that each dimensional constant we introduce enables us to remove a parameter, so that the nondimensional equation has fewer parameters than the dimensional one. This result on the reduction in the number of parameters from nondimensionalization is called the Buckingham Pi theorem. It is called the “pi” theorem because he used the symbols $\Pi_1, \Pi_2, \dots$ to represent the nondimensional groups.

Solution. The original dimensional equation

$$\dot{N} = rN \left(1 - \frac{N}{K}\right) - H$$

contains three parameters:

$$r, \quad K, \quad H.$$

After nondimensionalization, the equation is

$$\frac{dx}{d\tau} = x(1 - x) - \alpha,$$

which contains only one parameter:

$$\alpha = \frac{H}{rK}.$$

Therefore,

$$\boxed{\text{dimensional model: 3 parameters}}$$

while

$$\boxed{\text{nondimensional model: 1 parameter.}}$$

We introduced two dimensional scaling constants, T_0 and N_0 , and these allowed us to eliminate two of the original independent parameters.

Thus the parameter count has been reduced from

$$\boxed{3 \longrightarrow 1}.$$

Problem 2. Rotate your roles: questioner $\rightarrow$ timekeeper $\rightarrow$ scribe.

Let

$$\dot{N} = rN \left(1 - \frac{N}{K} \right) - \frac{HN}{A + N}.$$

This is a slightly different harvesting case.

(a) This harvesting term,

$$\frac{HN}{A + N},$$

is in the form of a special function called a Monod function. Plot an approximation of the Monod function by hand without using any plotting tools.

Solution. Consider the harvesting function

$$F(N) = \frac{HN}{A + N}.$$

At $N = 0$,

$$F(0) = 0.$$

At $N = A$,

$$F(A) = \frac{HA}{A + A} = \frac{H}{2}.$$

As $N \rightarrow \infty$,

$$\lim_{N \rightarrow \infty} \frac{HN}{A + N} = H.$$

Thus the horizontal asymptote is

$$\boxed{F(N) = H}.$$

For small N , we have

$$A + N \approx A,$$

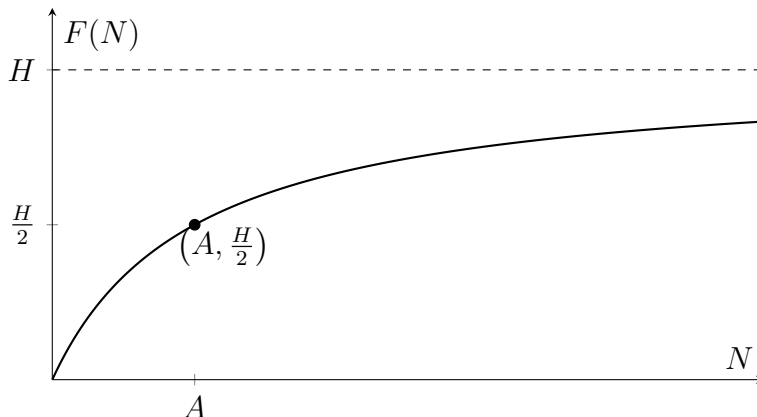
so

$$F(N) \approx \frac{H}{A}N.$$

Therefore, near $N = 0$, the function is approximately linear with slope

$$\frac{H}{A}.$$

A sketch of the function is:



The important qualitative features are therefore

$$F(0) = 0, \quad F(A) = \frac{H}{2}, \quad F(N) \rightarrow H \text{ as } N \rightarrow \infty.$$

- (b) What do you think of this function as a description of a harvesting process?

Solution. The function

$$F(N) = \frac{HN}{A + N}$$

gives a harvesting rate that depends on the available population.

When the population is small,

$$F(N) \approx \frac{H}{A}N,$$

so the harvesting rate is also small. Thus, when there are few individuals available, relatively few individuals are harvested.

As the population increases, the harvesting rate increases, but it does not increase without bound. Instead,

$$F(N) \rightarrow H.$$

Thus H represents the maximum harvesting rate.

This is a reasonable model when the ability to harvest is limited by the availability of the population at low population levels and by some maximum harvesting capacity at high population levels.

- (c) Identify the dimension of each of the variables and parameters. Once you nondimensionalize how many parameters do you expect to remain?

Solution. Again let P denote population and T denote time.

We have

$$[N] = P, \quad [t] = T.$$

As before,

$$[r] = T^{-1}, \quad [K] = P.$$

Because A is added to N ,

$$[A] = [N] = P.$$

Now consider

$$\frac{HN}{A + N}.$$

Since

$$\left[\frac{N}{A + N} \right] = 1,$$

the parameter H itself must have the dimension of population per unit time:

$$[H] = \frac{P}{T}.$$

Therefore,

$$[N] = P, \quad [t] = T, \quad [r] = T^{-1}, \quad [K] = P, \quad [A] = P, \quad [H] = PT^{-1}.$$

There are four dimensional parameters:

$$r, \quad K, \quad A, \quad H.$$

There are two fundamental dimensions involved:

$$P \quad \text{and} \quad T.$$

We therefore expect nondimensionalization to reduce the number of independent parameters by two. Hence we expect

$$\boxed{4 - 2 = 2}$$

independent nondimensional parameters to remain.

(d) Nondimensionalize this equation.

As part of this process, identify the dimensionless groups and check your work with another team.

There are multiple good choices for N_0 . What are some reasons to choose one or the other?

Solution. Introduce

$$x = \frac{N}{N_0}, \quad \tau = \frac{t}{T_0}.$$

Thus

$$N = N_0 x, \quad t = T_0 \tau.$$

As before,

$$\frac{dN}{dt} = \frac{N_0}{T_0} \frac{dx}{d\tau}.$$

Substituting into

$$\dot{N} = rN \left(1 - \frac{N}{K}\right) - \frac{HN}{A + N}$$

gives

$$\frac{N_0}{T_0} \frac{dx}{d\tau} = rN_0 x \left(1 - \frac{N_0 x}{K}\right) - \frac{HN_0 x}{A + N_0 x}.$$

Multiplying through by T_0/N_0 ,

$$\frac{dx}{d\tau} = rT_0 x \left(1 - \frac{N_0}{K}x\right) - \frac{HT_0 x}{A + N_0 x}.$$

Factor N_0 from the denominator of the harvesting term:

$$A + N_0 x = N_0 \left(\frac{A}{N_0} + x\right).$$

Therefore,

$$\boxed{\frac{dx}{d\tau} = rT_0 x \left(1 - \frac{N_0}{K}x\right) - \frac{HT_0}{N_0} \frac{x}{A/N_0 + x}.$$

The dimensionless groups that arise are

$$\boxed{rT_0, \quad \frac{N_0}{K}, \quad \frac{A}{N_0}, \quad \frac{HT_0}{N_0}.$$

These groups are not all independent because the freely chosen scales N_0 and T_0 can be used to eliminate two of them.

A convenient time scale is

$$T_0 = \frac{1}{r}.$$

For the population scale there are at least two especially natural choices.

Choice 1: $N_0 = K$. If

$$N_0 = K,$$

then $x = N/K$ measures population relative to the carrying capacity. The equation becomes

$$\frac{dx}{d\tau} = x(1 - x) - \frac{H}{rK} \frac{x}{A/K + x}.$$

Define

$$\alpha = \frac{H}{rK}, \quad \beta = \frac{A}{K}.$$

Then

$$\frac{dx}{d\tau} = x(1 - x) - \alpha \frac{x}{\beta + x}.$$

This choice is useful because the carrying capacity occurs at

$$x = 1.$$

Choice 2: $N_0 = A$. Alternatively, let

$$N_0 = A.$$

This is especially convenient for studying the Monod harvesting term. The equation becomes

$$\frac{dx}{d\tau} = x \left(1 - \frac{A}{K}x \right) - \frac{H}{rA} \frac{x}{1 + x}.$$

Define

$$\alpha = \frac{A}{K}, \quad \beta = \frac{H}{rA}.$$

Then

$$\frac{dx}{d\tau} = x(1 - \alpha x) - \beta \frac{x}{1 + x}.$$

This form contains only two nondimensional parameters,

$$\alpha = \frac{A}{K}, \quad \beta = \frac{H}{rA}.$$

Thus, as predicted, four dimensional parameters have been reduced to two nondimensional parameters.

Choosing $N_0 = K$ is convenient when the carrying capacity is the important population scale. Choosing $N_0 = A$ is convenient when the harvesting function is the primary object of interest, because the Monod function then takes the particularly simple form

$$\frac{x}{1 + x}.$$

- (e) Now that it is nondimensional, take another look at the expression for your harvesting function. Plot it using appropriate axis ticks. How did your axis tick labels change?

Solution. Using the choice

$$N_0 = A,$$

we have

$$x = \frac{N}{A}.$$

The original harvesting function is

$$F(N) = \frac{HN}{A+N}.$$

Divide the harvesting rate by its natural scale H . Define

$$f(x) = \frac{F(N)}{H}.$$

Since $N = Ax$,

$$f(x) = \frac{1}{H} \frac{H(Ax)}{A + Ax}.$$

Therefore,

$$f(x) = \frac{Ax}{A(1+x)} = \boxed{\frac{x}{1+x}}.$$

The dimensional function

$$F(N) = \frac{HN}{A+N}$$

has the important point

$$\left(A, \frac{H}{2}\right)$$

and horizontal asymptote

$$F = H.$$

After nondimensionalization these become

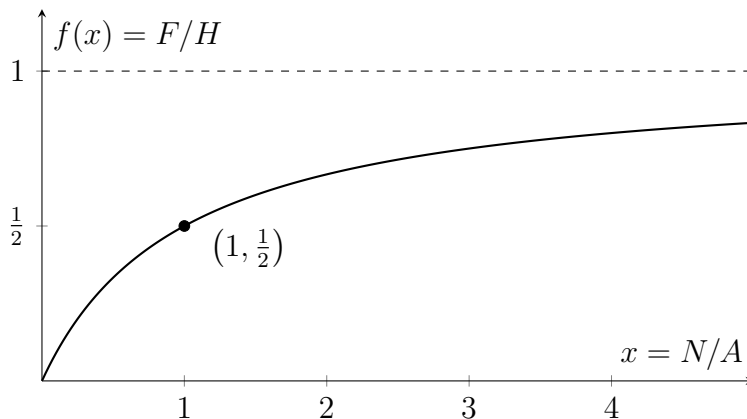
$$\boxed{\left(1, \frac{1}{2}\right)}$$

and

$$\boxed{f = 1}.$$

Thus all of the dimensional labels A and H disappear from the axes.

The nondimensional plot is:



Originally, increments on the horizontal axis were measured in units of A :

$$0, A, 2A, 3A, \dots$$

and vertical values were measured in units of H .

After nondimensionalization, the corresponding horizontal ticks are simply

$$0, 1, 2, 3, \dots$$

and the vertical asymptote value changes from the dimensional value H to the dimensionless value

$$1.$$

In particular,

$$\boxed{N = A \quad \longrightarrow \quad x = 1}$$

and

$$\boxed{F(A) = \frac{H}{2} \quad \longrightarrow \quad f(1) = \frac{1}{2}.$$